Notes: Card & Krueger (AER, 2000)

Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania: Reply

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1 Big picture

- **Question.** Does the April 1992 New Jersey minimum-wage increase from $4.25 to $5.05 reduce fast-food employment relative to nearby Pennsylvania, once the disputed payroll data and new administrative data are examined?
- **Context.** The paper is a reply to Neumark and Wascher's AER Comment, which argued that payroll-record data contradict Card and Krueger's original telephone-survey finding.
- **Main strategy.** Card and Krueger add a new representative payroll-based source: BLS ES-202 administrative records drawn from unemployment-insurance payroll-tax filings.
- **Central empirical message.** Representative data sets—the original Card-Krueger survey, the BLS ES-202 fast-food sample, and public BLS aggregate data—do not show statistically significant employment losses in New Jersey.
- **Key reconciliation.** The only sample producing a strongly negative New Jersey effect is the nonrepresentative Berman/EPI subset inside the BNW payroll sample.
- **Broader lesson.** The paper is about minimum wages, but also about replication standards: administrative data, sample frames, reporting intervals, and sensitivity analysis can reverse empirical conclusions.

2 Data and measurement

2.1 Policy event and comparison group

On April 1, 1992, New Jersey raised its state minimum wage from $4.25 to $5.05 per hour, while Pennsylvania remained at $4.25. The original study compared New Jersey restaurants to seven Pennsylvania border counties; this reply also uses a broader fourteen-county comparison region. The 1996 federal minimum-wage increase provides a reverse comparison because it was binding in Pennsylvania but not New Jersey.

Detailed interpretation. The design is a difference-in-differences comparison built around a state-specific policy change and nearby untreated areas.

2.2 BLS ES-202 administrative payroll data

The ES-202 data are employer-reported unemployment-insurance payroll-tax records. The monthly employment count is the number of covered workers employed during the pay period that includes the 12th day of the month. The longitudinal sample tracks a fixed set of reporting units belonging to large fast-food chains, and repeated cross sections cover late 1991 through 1997.

Detailed interpretation. This is the most important data contribution of the reply: the authors answer the "telephone survey" criticism using representative payroll data.

2.3 Original CK survey and BNW payroll sample

The original survey measured total restaurant employment before and after the New Jersey increase. BNW instead use payroll hours divided by 35 from a small sample composed partly of a Berman/EPI sample collected years after the treatment.

Detailed interpretation. The reply's main critique is not that payroll records are bad; it is that the BNW payroll sample is small, informally assembled, and potentially nonrepresentative.

2.4 Outcome variables and reporting interval concerns

The BLS and CK data measure total employees. BNW measure average payroll hours divided by 35. BNW restaurants report payroll hours over weekly, biweekly, or monthly intervals, which creates potential timing and seasonal biases.

Detailed interpretation. The measurement problem can be correlated with treatment status, so it can bias the New Jersey-Pennsylvania comparison.

3 Models and empirical specifications

3.1 Basic difference-in-differences regression

$$\Delta E_i = \alpha + \beta NJ_i + \gamma Chain_i + \delta Subunit_i + \varepsilon_i.$$

Detailed interpretation. The coefficient β measures the difference in employment growth between New Jersey and Pennsylvania after the 1992 minimum-wage increase.

3.2 Proportionate employment growth

$$g_i = \frac{E_{i,after} - E_{i,before}}{E_{i,before}}.$$

Detailed interpretation. This checks whether the results are driven by employment levels at larger restaurants.

3.3 BNW hours regressions

$$\Delta Hours_i/35 = \alpha + \beta NJ_i + \theta NWSubsample_i + Chain_i + Ownership_i + ReportingInterval_i + \varepsilon_i.$$

Detailed interpretation. Once chain, sample, ownership, and payroll-period controls are added, the New Jersey coefficient becomes smaller and statistically insignificant.

3.4 Within-New-Jersey wage-gap model

$$\Delta E_{cell} = \alpha + \beta GAP_{cell} + \varepsilon_{cell}.$$

Detailed interpretation. This uses only New Jersey variation and asks whether restaurants more affected by the minimum wage lost employment. They did not.

4 Identification and empirical design

The 1992 New Jersey minimum-wage increase identifies the primary treatment. The BLS ES-202 data are crucial because they are payroll-based and representative for the selected chains. The 1996 federal minimum-wage increase reverses treatment status by affecting Pennsylvania but not New Jersey. The BNW reanalysis tests whether the contrary result comes from sample composition and reporting intervals rather than from the use of payroll records.

5 Section-by-section study guide

- **Introduction:** sets up the replication dispute and the four main critiques of BNW.
- **Section I:** analyzes representative BLS fast-food payroll data and reports Tables 1–3.
- **Section II:** uses repeated BLS cross sections and the 1996 federal minimum-wage increase; Figure 2 is the key visual.
- **Section III:** reanalyzes BNW, focusing on sample representativeness, reporting intervals, and within-New-Jersey wage-gap evidence.
- **Conclusion:** synthesizes all evidence and concludes that the 1992 increase had little or no systematic effect on total fast-food employment.

6 Tables and figures

6.1 Figure 1: Areas of New Jersey and Pennsylvania Covered by Original Survey and BLS Data

Read: The map shows New Jersey, the original seven Pennsylvania counties, and the broader fourteen-county Pennsylvania comparison region.

Detailed interpretation. The figure explains why the paper reports both narrow and broad Pennsylvania controls.

Economic message. The map frames the empirical treatment-control comparison.

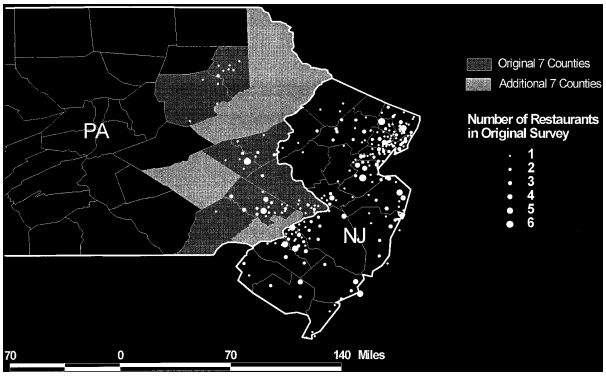


Table 1—Descriptive Statistics for Fast-Food Restaurants Drawn from BLS ES-202 Data and Card-Krueger Survey

	Means with standard deviations in parentheses:								
	New Jersey			7 Pennsylvania counties			14 Pennsylvania counties		
	Before	After	Change	Before	After	Change	Before	After	Change
A. *BLS ES-202 Data*									
February–March 1992 to November–December 1992	37.2	37.6	0.41	42.5	42.4	−0.12	44.8	44.3	−0.53
	(19.9)	(21.0)	(9.82)	(23.2)	(23.5)	(10.94)	(53.7)	(59.9)	(12.32)
February 1992 to November 1992	37.2	37.8	0.57	42.7	42.2	−0.54	44.9	44.4	−0.58
	(19.9)	(20.9)	(10.12)	(23.8)	(23.2)	(12.82)	(53.6)	(60.4)	(13.83)
March 1992 to March 1993	37.2	34.8	−2.48	42.3	37.5	−4.80	44.7	40.7	−4.0
	(20.1)	(20.0)	(13.99)	(22.8)	(18.6)	(22.74)	(54.0)	(54.5)	(18.1)
B. *Card-Krueger Survey Data*									
February 1992 to November 1992	29.8	30.0	0.19	33.1	30.9	−2.23	NA	NA	NA
	(12.5)	(13.0)	(9.82)	(14.7)	(10.6)	(11.98)			

Notes: Sample sizes for the first two rows are 437 for New Jersey, 127 for Pennsylvania 7 counties, and 250 for Pennsylvania 14 counties; sample sizes for third row are 436, 127, and 250, respectively; sample sizes for the last row are 309 for New Jersey and 75 for Pennsylvania. The 7 Pennsylvania counties used in the middle columns are the same counties used in Card and Krueger (1994); these 7 counties are a subset of the 14 counties in the last three columns (see text). The unit of observation for the BLS data is the "reporting unit," which in some cases includes multiple establishments. The unit of observation in the Card-Krueger data is the individual restaurant.

6.2 Table 1: Descriptive Statistics for Fast-Food Restaurants Drawn from BLS ES-202 Data and Card-Krueger Survey

Read: BLS data show New Jersey employment rising slightly while Pennsylvania falls over the main survey window. The CK survey shows a similar pattern.

Detailed interpretation. The raw means already undermine the claim that payroll data necessarily show New Jersey losses.

Economic message. Representative payroll records look qualitatively similar to the original CK survey.

6.3 Table 2: Basic Regression Results; BLS ES-202 Fast-Food Data and Card-Krueger Survey Data

Read: BLS coefficients on the New Jersey indicator are small and generally positive; CK survey coefficients are also positive.

Detailed interpretation. The table is the main regression analogue of the descriptive statistics.

Economic message. Clean payroll-record evidence and the original survey evidence tell the same story.

Table 2—Basic Regression Results; BLS ES-202 Fast-Food Data and Card-Krueger Survey Data

	Dependent variable:			
	Change in levels		Proportionate change	
Explanatory variables	(1)	(2)	(3)	(4)
A. *All of New Jersey and 7 Pennsylvania Counties, BLS Data*				
New Jersey indicator	0.536	0.225	0.007	0.009
	(1.017)	(1.029)	(0.029)	(0.029)
Chain dummies and subunit dummy variable	No	Yes	No	Yes
Standard error of regression	10.09	9.99	0.286	0.281
R^2	0.001	0.029	0.000	0.046
B. *All of New Jersey and 14 Pennsylvania Counties, BLS Data*				
New Jersey indicator	0.946	0.272	0.045	0.032
	(0.856)	(0.859)	(0.024)	(0.024)
Chain dummies and subunit dummy variable	No	Yes	No	Yes
Standard error of regression	10.80	10.63	0.303	0.294
R^2	0.002	0.042	0.005	0.071
C. *Original Card-Krueger Survey Data*				
New Jersey indicator	2.411	2.488	0.029	0.030
	(1.323)	(1.323)	(0.050)	(0.049)
Chain and company-ownership dummies	No	Yes	No	Yes
Standard error of regression	10.28	10.25	0.385	0.382
R^2	0.009	0.025	0.001	0.024

Notes: Each regression also includes a constant. Sample size is 564 for panel A, 687 for panel B, and 384 for panel C. Subunit dummy variable equals one if the reporting unit is a subunit of a multiunit employer. For comparability with the BLS data, employment in the CK sample is measured by the total number of full- and part-time employees. Standard errors are in parentheses.

Table 3—Sensitivity of New Jersey Employment Growth Differential to Specification Changes; BLS ES-202 Fast-Food Restaurant Sample

Specification and sample	Change in levels (1)	Proportionate change (2)	Sample size
A. New Jersey and 7 Pennsylvania Counties			
1. Basic specification	0.225	0.009	564
	(1.029)	(0.029)	
2. Excluding closed stores	0.909	0.031	549
	(0.950)	(0.024)	
3. Excluding closed stores unless imputation code = 9	0.640	0.022	553
	(0.973)	(0.025)	
4. Drop large outlier	0.251	0.009	563
	(0.970)	(0.028)	
5. Proportionate change with initial employment in base	—	−0.001	564
		(0.032)	
6. Excluding New Jersey shore	0.032	0.008	480
	(1.092)	(0.030)	
7. March 1992 to March 1993 employment change	2.345	0.007	563
	(1.678)	(0.035)	
8. February 1992 to November 1992 employment change	1.05	0.013	564
	(1.10)	(0.032)	
B. New Jersey and 14 Pennsylvania Counties			
1. Basic specification	0.272	0.032	687
	(1.029)	(0.024)	
2. Excluding closed stores	0.639	0.055	671
	(0.776)	(0.021)	
3. Excluding closed stores unless imputation code = 9	0.338	0.044	675
	(0.787)	(0.021)	
4. Drop large outliers	0.72	0.032	685
	(0.78)	(0.023)	
5. Proportionate change with initial employment in base	—	0.020	687
		(0.024)	
6. Excluding New Jersey shore	0.069	0.030	603
	(0.924)	(0.025)	
7. March 1992 to March 1993 employment change	1.196	0.009	686
	(1.258)	(0.027)	
8. February 1992 to November 1992 employment change	0.624	0.027	687
	(0.927)	(0.024)	

Notes: The table reports the coefficient (with standard error in parentheses) for the New Jersey dummy variable from a regression of the change in employment [column (1)] or proportionate change in employment [column (2)] on a New Jersey dummy variable, chain dummy variables, a dummy variable indicating whether the restaurant is reported as a subunit of a multiestablishment employer, and a constant.

6.4 Table 3: Sensitivity of New Jersey Employment Growth Differential to Specification Changes; BLS ES-202 Fast-Food Restaurant Sample

Read: The table varies closed-store treatment, outliers, shore counties, time windows, and Pennsylvania controls. Estimates remain near zero or positive.

Detailed interpretation. Reasonable specification changes do not produce statistically significant negative effects.

Economic message. If there were a large employment loss, it should appear under some reasonable BLS specification.

6.5 Figure 2: Employment in New Jersey and Pennsylvania Fast-Food Restaurants, October 1991 to September 1997

Read: The figure shows indexed employment for New Jersey, seven Pennsylvania counties, and fourteen Pennsylvania counties from 1991 to 1997.

Detailed interpretation. The long series extends the evidence beyond the short survey window and uses the 1996 federal increase as a reverse experiment.

Economic message. There is little evidence that either minimum-wage increase reduced fast-food employment.

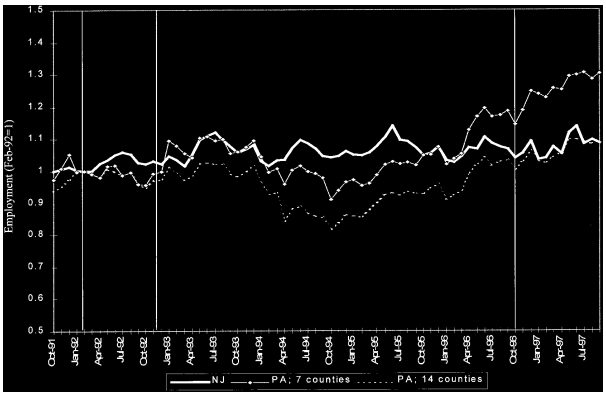


Table 4—Descriptive Statistics for Levels and Changes in Employment by State, BNW Data

	Means with standard deviations in parentheses:						
	New Jersey			Pennsylvania			Difference-in-differences New Jersey-Pennsylvania (standard error)
	Before	After	Change	Before	After	Change	
Total payroll hours/35:							
1. Pooled BNW sample	17.5	17.5	−0.1	15.1	15.9	0.8	−0.85
	(5.5)	(5.9)	(3.4)	(4.0)	(5.9)	(3.5)	(0.49)
2. NW subsample	17.7	16.7	−1.0	13.4	12.4	−1.0	−0.05
	(6.1)	(6.3)	(3.3)	(3.8)	(4.9)	(3.5)	(0.61)
3. Original Berman subsample	17.1	19.3	2.1	16.9	20.4	3.4	−1.28
	(3.5)	(4.3)	(2.7)	(3.4)	(4.3)	(2.1)	(0.63)
Nonmanagement employment:							
4. Pooled BNW sample	24.8	28.4	3.6	29.0	31.3	2.2	1.39
	(6.0)	(6.8)	(3.0)	(5.5)	(6.8)	(4.7)	(1.20)

Notes: See text for description of employment variables and samples. Sample sizes are as follows. Row 1: New Jersey 163; Pennsylvania 72. Row 2: New Jersey 114; Pennsylvania 40. Row 3: New Jersey 49; Pennsylvania 23. Row 4: New Jersey 19; Pennsylvania 33.

6.6 Table 4: Descriptive Statistics for Levels and Changes in Employment by State, BNW Data

Read: BNW uses average payroll hours divided by 35 and displays different patterns across subsamples.

Detailed interpretation. The table introduces the BNW sample-composition problem.

Economic message. The negative finding comes from a small, nonrepresentative payroll sample, not from payroll data as a class.

6.7 Table 5: Estimated Regression Models for Change in Average Payroll Hours/35, BNW Data

Read: The simple New Jersey coefficient is negative, but controls for subsample, chain, ownership, and reporting interval make it smaller and insignificant.

Detailed interpretation. Payroll hours are sensitive to reporting intervals and sample composition.

Economic message. Once measurement factors are controlled, BNW no longer provides strong evidence of New Jersey employment decline.

Table 5—Estimated Regression Models for Change in Average Payroll Hours/35, BNW Data

	Specification:						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
New Jersey	−0.85	—	—	—	−0.36	−0.66	−0.09
	(0.49)				(0.44)	(0.41)	(0.42)
NW subsample (1 = yes)	—	−3.49	—	—	−3.44	—	—
		(0.42)			(0.43)		
Chain dummies:							
Roy Rogers	—	—	−3.56	—	—	−3.14	−1.98
			(0.81)			(0.85)	(0.89)
Wendy's	—	—	−0.85	—	—	−0.71	−1.35
			(0.67)			(0.67)	(0.70)
KFC	—	—	−6.51	—	—	−6.30	−6.56
			(0.90)			(0.90)	(0.89)
Company-owned	—	—	−0.89	—	—	−1.31	−0.72
			(0.76)			(0.81)	(0.95)
Payroll data type:							
Biweekly	—	—	—	1.73	—	—	1.65
				(0.52)			(0.52)
Monthly	—	—	—	−2.60	—	—	−1.06
				(0.48)			(0.89)
R^2	0.01	0.23	0.41	0.30	0.23	0.10	0.45
Standard error of regression	3.47	3.07	2.70	2.95	3.08	3.32	2.62

Notes: Standard errors are in parentheses. Sample consists of 235 stores. Dependent variable in all models is the change in average weekly payroll hours divided by 35 between wave 1 and wave 2.

Table 6—Estimated Relative Employment Effects in New Jersey for Alternative Samples and Specifications, BNW Data

	Pooled BNW sample	Neumark-Wascher sample	Original Berman sample
A. *Change in Average Payroll Hours/35*			
1. No controls	−0.85	−0.05	−1.28
	(0.49)	(0.61)	(0.63)
2. Exclude first Pennsylvania franchisee, no controls	0.37	−0.11	—
	(0.56)	(0.62)	
3. Controls for chain and ownership	−0.66	−0.27	−0.95
	(0.41)	(0.53)	(0.64)
4. Controls for chain, ownership and payroll period	−0.09	−0.22	0.58
	(0.42)	(0.54)	(0.72)
B. *Change in Payroll Hours/35 Using First Pay Period per Restaurant*			
5. No controls	−0.55	0.18	−0.85
	(0.50)	(0.63)	(0.67)
6. Controls for chain, ownership, and payroll period	0.07	−0.15	0.96
	(0.43)	(0.52)	(0.78)
C. *Proportional Change in Average Payroll Hours/35*			
7. No controls	−0.06	−0.00	−0.09
	(0.05)	(0.07)	(0.07)
8. Controls for chain and ownership	−0.05	−0.03	−0.04
	(0.05)	(0.07)	(0.07)
9. Controls for chain, ownership, and payroll period	−0.01	−0.03	−0.08
	(0.05)	(0.07)	(0.08)

Notes: Pooled BNW sample has 235 observations; NW sample has 154 observations; original Berman sample has 71 observations. In row 2, data for 26 restaurants owned by one franchisee are excluded. In this row only, pooled BNW sample has 209 observations; and NW sample has 151 observations. Dependent variable in panel A is the change in average payroll hours between the first and second waves, divided by 35. Dependent variable in panel B is the change in payroll hours for the first payroll period reported by each store between the first and second waves, divided by 35. Dependent variable in panel C is the change in average payroll hours between the first and second waves, divided by average payroll hours in the first and second waves. Standard errors are in parentheses.

6.8 Table 6: Estimated Relative Employment Effects in New Jersey for Alternative Samples and Specifications, BNW Data

Read: Alternative sample restrictions and model specifications shift the New Jersey coefficient.

Detailed interpretation. The negative BNW result is fragile and depends on sample construction and reporting choices.

Economic message. A robust causal effect should not depend heavily on a single franchisee or reporting convention.

6.9 Table 7: Comparisons of Employment Growth by Chain and Zip-Code Area, Card-Krueger Data versus BNW Data

Read: Cell-level CK and BNW changes line up better in New Jersey than in Pennsylvania.

Detailed interpretation. The discrepancy is concentrated in Pennsylvania BNW observations.

Economic message. The conflict is not primarily about New Jersey restaurants; it is about the Pennsylvania BNW sample.

Table 7—Comparisons of Employment Growth by Chain and Zip-Code Area, Card-Krueger Data Versus Berman-Neumark-Wascher Data

	New Jersey and Pennsylvania	New Jersey only	Pennsylvania only
A. *Using Combined BNW Sample*			
Intercept	−0.32	0.41	−3.91
	(0.56)	(0.50)	(1.77)
Change in employment in BNW sample	0.78	0.90	0.65
	(0.22)	(0.19)	(0.68)
R^2	0.22	0.38	0.09
Standard error	8.97	7.35	10.76
p-value: intercept = 0, slope = 1	0.47	0.65	0.07
Number of observations (chain × zip-code cells)	48	37	11
B. *Using Combined BNW Sample Excluding Data from One Franchisee*			
Intercept	−0.26	0.36	−3.52
	(0.54)	(0.51)	(1.67)
Change in employment in BNW sample	0.87	0.91	0.93
	(0.21)	(0.20)	(0.73)
R^2	0.28	0.39	0.17
Standard error	8.56	7.40	10.27
p-value: intercept = 0, slope = 1	0.71	0.72	0.14
Number of observations (chain × zip-code cells)	46	36	10

Notes: Dependent variable in all models is the mean change in full-time employment for fast-food restaurants of a specific chain in a specific three-digit zip-code area, taken from the Card-Krueger data set. Independent variable is the mean change in payroll hours divided by 35 for fast-food restaurants (in the same chain and zip-code area) taken from the BNW data set. In panel B, restaurants in the BNW sample obtained from the franchisee who provided Berman's Pennsylvania data are deleted prior to forming average employment changes by chain and zip-code area. All models are fit by weighted least squares using as a weight the number of observations in the chain-by-zip-code cell in the Card-Krueger data set. Standard errors are in parentheses.

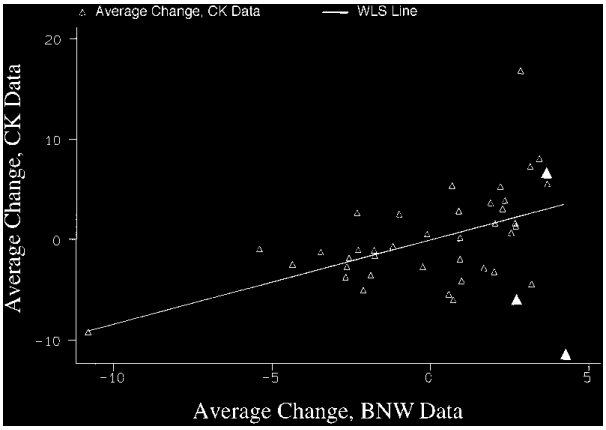
Figure 1: Table 7: CK versus BNW cell-level comparison

6.10 Figure 3 and Table 8: CK Data versus BNW Data and BNW versus BLS ES-202 Data

Read: Figure 3 shows CK and BNW cell-level changes. Table 8 relates BNW changes to BLS cell-level changes and EPI sample shares.

Detailed interpretation. Berman/EPI Pennsylvania cells are influential and appear nonrepresentative.

Economic message. The EPI Pennsylvania data are the key anomaly generating the negative relative trend.



(a) Figure 3: CK versus BNW

Table 8—Comparisons of Employment Growth by Chain and Zip-Code Area, Berman-Neumark-Wascher Data Versus Bureau of Labor Statistics ES-202 Data (Dependent Variable: Average Change in Employment, BNW Data)

	(1)	(2)	(3)
Average change in employment, BLS data	0.17	0.10	0.09
	(0.05)	(0.05)	(0.05)
Fraction of sample collected by EPI	—	3.46	—
		(0.70)	
Fraction of sample collected by EPI × Pennsylvania	—	—	4.33
			(1.19)
Fraction of sample collected by EPI × New Jersey	—	—	3.22
			(0.75)
Constant	0.04	−0.95	−0.96
	(0.34)	(0.34)	(0.34)
R^2	0.19	0.49	0.50

Notes: Weighted least-squares estimates are presented. Weights are equal to the number of BNW observations in the cell. Cells are composed of the three-digit zip-code-by-chain areas. Standard errors are in parentheses.

(b) Table 8: BNW versus BLS

6.11 Figure 4: Average FTE Employment in Low- and High-Wage Areas of New Jersey, Before and After 1992 Minimum-Wage Increase

Read: Low-wage areas were more affected by the minimum wage and show faster employment growth than high-wage areas.

Detailed interpretation. This within-New-Jersey evidence avoids the Pennsylvania control problem.

Economic message. More affected restaurants did not lose employment relative to less affected restaurants.

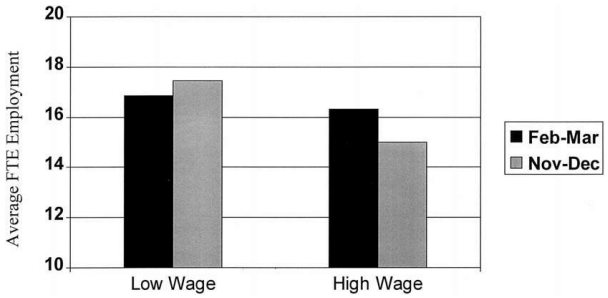


Table 9—Estimates of Minimum-Wage Effects on Employment Growth in the Eating and Drinking Industry from February/March to November/December

	BLS 790 data, New Jersey and Pennsylvania			ES-202 data, New Jersey and 7 counties of Pennsylvania	
	NW (1)	Revised data (2)	Revised data and correctly dated unemployment (3)	NW (4)	Correctly dated unemployment (5)
Minimum-wage change	−0.15	−0.11	−0.03	−0.11	−0.01
	(0.10)	(0.10)	(0.08)	(0.10)	(0.07)
New Jersey	−0.05	−0.38	−0.41	0.25	0.21
	(0.95)	(0.95)	(0.80)	(0.96)	(0.72)
Change in unemployment[a]	−0.32	−0.39	−1.47	−0.49	−1.84
	(0.47)	(0.46)	(0.44)	(0.46)	(0.39)
R^2	0.15	0.13	0.38	0.14	0.51

Notes: Standard errors are in parentheses. Models are estimated using changes in employment in SIC 58 (eating and drinking) between February/March and November/December of 1982–1996. Dependent variable in columns (1)–(3) is the percentage change in statewide employment in SIC 58 from the BLS 790 program. Dependent variable in columns (4)–(5) is the percentage change in employment from the ES202 program; New Jersey data are statewide and Pennsylvania data are for 7 counties only. Columns (1) and (4) are taken from Neumark and Wascher (2000 Table 10). Models in columns (2)–(3) use revised BLS 790 data for 1995 and 1996. Sample size is 30 observations.

[a] In columns (1), (2), and (4) the change in unemployment represents the change in the annual average unemployment rate. In columns (3) and (5) the change in unemployment represents the difference between the average of the seasonally adjusted rates in February and March and the average of the seasonally adjusted rates in the following November and December.

6.12 Table 9: Estimates of Minimum-Wage Effects on Employment Growth in the Eating and Drinking Industry

Read: Replacing unrevised data and correcting controls weakens Neumark-Wascher's negative aggregate finding.

Detailed interpretation. The aggregate BLS result is also fragile.

Economic message. Public aggregate data are consistent with the representative fast-food evidence.

7 Short summary

- The reply evaluates Neumark and Wascher's claim that payroll records overturn Card and Krueger's original result.
- Representative BLS ES-202 payroll records show no significant negative New Jersey employment effect.
- Robustness checks and repeated cross sections also show little evidence of job loss.
- The BNW negative result is tied to nonrepresentative sample construction and reporting interval issues.
- Within New Jersey, low-wage areas most affected by the minimum wage grew faster than high-wage areas.
- The authors conclude that the 1992 increase had little or no systematic effect on total fast-food employment.

8 Conclusion

8.1 Core conclusion

Card and Krueger conclude that the April 1992 New Jersey minimum-wage increase probably had no systematic negative effect on total fast-food employment.

8.2 What the reply adds

The reply adds administrative BLS payroll data, a broader Pennsylvania comparison region, longer time-series evidence, a reverse 1996 minimum-wage comparison, and a detailed audit of BNW.

8.3 Interpretation of the replication debate

The issue is not telephone survey versus payroll record. BLS ES-202 is payroll-based and supports the original conclusion. The issue is sample representativeness and measurement.

8.4 Economic intuition

A modest minimum wage may not reduce employment if labor-market frictions, monopsony power, reduced turnover, or firm heterogeneity offset job cuts.

8.5 Policy implication

The evidence does not support a strong claim that the 1992 New Jersey increase reduced fast-food employment. The conclusion is specific to this sector, region, and policy size.

8.6 Methodological takeaway

Empirical conclusions can depend on sampling frames, post hoc data collection, reporting intervals, and control alignment. Representative administrative data are valuable for replication.

8.7 Final takeaway

The best available evidence in the debate does not support a negative total-employment effect from the New Jersey minimum-wage increase.